

SS1 Science Track: Chemistry Diagnostic (Part 2)

Scope: Acids, Bases, Energy, Rates & Organic Chemistry
Questions 21 - 40

Section B: Acids, Bases and Salts

No.	Question & Options	Ans	Diagnostic Focus
21	A solution has a pH of 2. This solution is: A) Strongly acidic B) Weakly acidic C) Strongly alkaline D) Neutral	A	pH Scale understanding.
22	Which of the following is a characteristic property of bases? A) Turn blue litmus paper red B) React with acids to form salt and water C) Have a sour taste D) React with carbonates to release CO_2	B	Properties of Bases.
23	When dilute hydrochloric acid reacts with Calcium Carbonate, the gas evolved is: A) Hydrogen B) Oxygen C) Carbon Dioxide D) Chlorine	C	Acid Reactions (Carbonates).
24	Which salt can be prepared by the method of Precipitation ? A) Sodium Chloride B) Barium Sulfate C) Potassium Nitrate D) Copper (II) Sulfate	B	Salt Preparation (Solubility).
25	An amphoteric oxide is one that reacts with: A) Acids only B) Bases only C) Both acids and bases D) Water only	C	Oxide Classification.

Section C: Energy Changes, Rates & RedOx

No.	Question & Options	Ans	Diagnostic Focus
26	In an exothermic reaction, the temperature of the surroundings: A) Stays the same B) Decreases C) Increases D) Drops to zero	C	Energy Changes (Enthalpy).
27	According to Collision Theory, increasing the temperature increases the rate of reaction because: A) Particles get larger B) Particles collide more frequently and with more energy C) Activation energy is lowered D) Concentration increases	B	Rates of Reaction (Theory).
28	A catalyst speed up a reaction by: A) Being consumed B) Providing an alternative path with lower activation energy C) Increasing the pressure D) Increasing the yield of product	B	Function of Catalysts.
29	In the reaction: $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$, which substance is reduced? A) CuO B) H_2 C) Cu D) H_2O	A	RedOx (Oxidation/Reduction).
30		B	OIL RIG (Electron transfer).

No.	Question & Options	Ans	Diagnostic Focus
	Oxidation is the: A) Gain of electrons B) Loss of electrons C) Loss of oxygen D) Gain of hydrogen		

Section D: Metals & Organic Chemistry Introduction

No.	Question & Options	Ans	Diagnostic Focus
31	Which metal is extracted from its ore using electrolysis due to its high reactivity? A) Iron B) Copper C) Aluminum D) Silver	C	Reactivity Series/Extraction.
32	Hematite is an important ore of which metal? A) Zinc B) Iron C) Lead D) Tin	B	Ores and Extraction.
33	Bronze is an alloy of Copper and: A) Zinc B) Tin C) Lead D) Iron	B	Alloys.
34	The main constituent of Natural Gas is: A) Ethane B) Methane C) Propane D) Butane	B	Organic Chemistry (Alkanes).
35	What is the general formula for the Alkane homologous series? A) C_nH_{2n} B) C_nH_{2n+2} C) C_nH_{2n-2} D) $C_nH_{2n+1}OH$	B	Homologous Series identification.
36	Fractional distillation of crude oil separates it based on differences in: A) Solubility B) Density C) Boiling Point D) Reactivity	C	Refining Petroleum.
37	Which gas is a major contributor to global warming? A) Nitrogen B) Argon C) Carbon Dioxide D) Helium	C	Environmental Chemistry.
38	An unsaturated hydrocarbon like Ethene will turn aqueous Bromine from orange-brown to: A) Blue B) Colourless C) Green D) Purple	B	Chemical Tests (Saturation).
39	Which of these is a synthetic polymer? A) Starch B) Cellulose C) Nylon D) Protein	C	Polymers/Macromolecules.
40	Diagnostic Logic: High scores in Section B/C indicate readiness for Pure Science. Struggles in D suggest difficulty with nomenclature and applied industrial chemistry.	-	Pathway Insight.